Fermionic Matter and Chirality from Projective Dirac Admissibility

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Abstract

The preceding gauge–gravity synthesis of the Cosmochrony programme showed that gravity and Yang–Mills dynamics arise from the a_2 and a_4 Seeley–DeWitt responses of the same admissible spectral functional. The present paper extends this principle to the fermionic sector. Fermions are not introduced as external matter fields. They arise as the spinorial face of the Weil module already forced by non-injective admissibility, once its metaplectic Lie-algebraic structure is complexified by the Lorentzian metric established in the geometric branch of the programme.

Three structural results are proved. First, the admissible Weil module V_ρ induces, through its metaplectic lift and the Lorentzian complexification $\mathfrak{mp}(2,\mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$, an admissible spinor bundle S_Π whose symmetric and determinant tensor sectors reproduce the $\mathrm{SU}(2)_L \times U(1)_Y$ electroweak bundle structure. Second, the projected Dirac operator $\mathcal{D}_{\Pi,g,A}$ contains a canonical zero-order endomorphism E_Π, the internal spectral residue of non-injective projection. Under the spinorial lift of BI parity (Theorem 3.7), E_Π is left-admissible and produces the $V-A$ chiral structure. The γ_5-weighted a_4 coefficient of $\mathcal{D}^2_{\Pi,g,A}$ then imposes anomaly-cancellation trace constraints on the hypercharge weights, which are thereby not free parameters but spectral coherence data. Third, the admissible saturation invariant $\sigma_c(n_3) = 3$ admits a spinorial multiplicity reading, distinct from its geometric reading as three spatial directions, yielding a gauge-singlet three-generation factor $\mathbb{C}^3_{\mathrm{gen}} \subset \ker(\mathrm{ad}_{\mathrm{SU}(2)} \oplus Y)$. The colour-coupled quark sector is obtained by tensoring with the colour module V_{color}, unconditional at the pointwise level ($[H\text{-color}]_{\mathrm{pointwise}}$ established in [15]). Finally, we identify a qualitative mechanism by which the oriented cascade lifts the static J_Π-protected degeneracy of the generation factor; the amplitude of this splitting remains deferred to the cascade normalisation.

Keywords: admissible spinor bundle; metaplectic lift; non-injective projection; Weil representation; projected Dirac operator; chiral selection; hypercharge rigidity; generation multiplicity; spectral anomaly cancellation; Cosmochrony

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1 Introduction

1.1 Position of the problem

The preceding papers of the programme establish the bosonic spectral stratification of admissible non-injective projection. The Gravity trilogy [5, 18, 14] identifies the Seeley–DeWitt level as the organising principle of emergent interactions: the horizontal a_2 response yields the Einstein sector, the vertical a_4 response yields the Yang–Mills sector, and the a_6 coefficient encodes the leading mixed gauge–gravity coupling. What lies outside this bosonic synthesis is the fermionic sector: the structural origin of spinors, chirality, the distribution of hypercharge, and the generation multiplicity.

The purpose of the present paper is to show that fermions need not be introduced as external matter fields added to the admissible bundle. They arise as the spinorial face of the Weil module already forced by non-injective admissibility, once its metaplectic lift is interpreted through the Lorentzian geometry closed in the geometric branch of the programme [16].

1.2 Main structural chain

The structural chain studied in this paper is

$$\Pi \;\Rightarrow\; F_n \simeq V_\rho \;\Rightarrow\; \mathrm{Mp}(2,\mathbb{R}) \;\Rightarrow\; \mathfrak{mp}(2,\mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C}) \;\rightsquigarrow\; \mathrm{Spin}(3,1) \simeq \mathrm{SL}(2,\mathbb{C}) \;\Rightarrow\; S_\Pi. \quad (1)$$

The identification $F_n \simeq V_\rho$ is the Weil-module characterisation of the admissible fibre, established from axioms A1–A3 and the Stone–von Neumann theorem in [1, 6]. The passage to $\mathrm{Mp}(2,\mathbb{R})$ is the metaplectic lift of the symplectic structure canonically carried by V_ρ. The Lorentzian closure of the effective metric forces the complexification $\mathfrak{mp}(2,\mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$, whose Lorentzian integration is the spin group $\mathrm{Spin}(3,1) \simeq \mathrm{SL}(2,\mathbb{C})$, yielding the admissible spinor bundle S_Π.

The tensor functors of S_Π then define the electroweak bundle data:

$$\mathrm{Sym}^2(S_\Pi) \;\Rightarrow\; \mathrm{ad}_{\mathbb{C}}(P_\Pi), \qquad \wedge^2(S_\Pi) \;\Rightarrow\; L_Y. \quad (2)$$

The symmetric sector gives the complex adjoint; after selecting the compact real form, it defines the $\mathrm{SU}(2)_L$ gauge sector. The determinant line $L_Y = \wedge^2(S_\Pi)$ is the geometric support of the $U(1)_Y$ factor. The distribution of hypercharge weights Y_R is not assumed: it is constrained by requiring that the projected spectral functional $S_\Pi[g, A, \mathcal{D}_{\Pi,g,A}]$ be invariant under $U(1)_Y$ gauge transformations.

1.3 Three modular results

The paper is organised around three independent but compatible results.

Theorem A (Spinorial electroweak bundle). The admissible Weil module V_ρ induces, via the metaplectic lift and Lorentzian complexification, a $\mathrm{Spin}(3,1)$ spinor bundle S_Π. Its symmetric and determinant tensor sectors reproduce the $\mathrm{SU}(2)_L \times U(1)_Y$ electroweak bundle structure (equations (2)). This result is algebraic and functorial; it depends on the Lorentzian closure of [16] for the complexification step.

Theorem B (Chiral selection and hypercharge rigidity). The projected Dirac operator satisfies the Lichnerowicz-type identity

$$\mathcal{D}^2_{\Pi,g,A} = -(\nabla^{S,A})^2 + \frac{R}{4} + \frac{1}{2}\gamma^\mu\gamma^\nu F^a_{\mu\nu} T^a_R + E_\Pi, \quad (3)$$

where E_Π is the internal spectral residue of non-injective projection in the spinorial sector. The term E_Π is responsible for chiral selection $P_L S_\Pi$, for the $V-A$ structure of weak couplings, and for the spectral coherence conditions constraining the hypercharge assignments Y_R via anomaly cancellation. This result is spectral and variational; it depends on the gauge sector of [4, 18] and on the gauge–gravity synthesis of [14] for the coupling of the Dirac sector to the joint spectral geometry.

Theorem C (Generation multiplicity). The admissible saturation invariant $\sigma_c(n_3) = 3$ [11] admits two functorially distinct readings of the same rank-three admissible invariant. In the geometric branch it yields the three effective spatial directions (established in [2, 16]). In the spinorial branch it yields a gauge-singlet generation space

$$\mathbb{C}^3_{\text{gen}} \;\subset\; \ker\bigl(\mathrm{ad}_{\mathrm{SU}(2)} \oplus Y\bigr)\,, \tag{4}$$

which is not identified with $\mathrm{Sym}^2(S_\Pi)$. The three-generation structure is a separate functorial image of the same admissible rank-three invariant, and $\mathbb{R}^3_{\text{space}} \neq \mathbb{C}^3_{\text{gen}}$ is established explicitly in Section 5. This result is combinatorial and representational; it depends on [11].

1.4 Status and dependencies

The full matter content targeted by the construction is

$$S_\Pi^{\text{matter}} = \bigl(S_\Pi \oplus (S_\Pi \otimes V_{\text{color}})\bigr) \otimes \mathbb{C}^3_{\text{gen}}, \tag{5}$$

where $V_{\text{color}} \in \mathrm{Rep}(\mathrm{SU}(3))$. The following status distinctions apply throughout the paper. The electroweak spinorial sector is structural once Theorem A is established. The colour-coupled quark bundle $S_\Pi \otimes V_{\text{color}}$ is unconditional at the pointwise level: $[H\text{-color}]_{\text{pointwise}}$ (exact equality at finite q) and $[H\text{-color}]_{\text{eff}}$ are established analytically in [15, 13]. The hypercharge rigidity of Theorem B and the generation multiplicity of Theorem C are proved below; they are not assumed as inputs.

The spinorial lift of BI parity, formerly the only additional technical condition of the fermionic sector, is established as Theorem 3.7. The $V-A$ structure and the hypercharge rigidity of Theorem B are therefore unconditional, their only remaining dependence being the axiomatic chain A1–A4 [1] and the upstream BI parity datum [12, 17].

This paper depends simultaneously on four layers of the programme: the admissible Weil fibre structure (Foundation, HeisenbergStructure, O-series); the closed Lorentzian geometry (Q5b through Q11); the gauge connection (Q6a, Q12); and the gauge–gravity synthesis (Q13, Gravity 3.0). It is accordingly the first paper of the Q-series whose construction depends on the simultaneous use of all four layers.

1.5 Organisation

Section 2 constructs the metaplectic lift and the admissible spinor bundle S_Π, proving Theorem A. Section 3 defines the projected Dirac operator, identifies E_Π as the spectral residue of non-injective projection in the spinorial sector, and proves the chiral selection result of Theorem B. Section 4 derives the hypercharge rigidity from spectral anomaly cancellation. Section 5 establishes the spinorial multiplicity reading of $\sigma_c(n_3) = 3$ and proves Theorem C. Section 6 studies the first structural constraint on the spectrum of E_Π^2 on the generation factor: a static J_Π-real, weight-preserving restriction cannot split the outer pair, whereas the oriented cascade generator carries a non-zero J_3 component capable of lifting this degeneracy. Section 7 discusses the colour-coupled quark sector and the pointwise analytical status of $[H\text{-color}]$. Section 8 summarises the results and identifies the remaining open directions.

2 Metaplectic Lift and the Admissible Spinor Bundle

2.1 From the Weil module to the metaplectic lift

The admissible fibre F_n has been identified with the Weil module V_ρ of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ from the axioms A1–A3 and the Stone–von Neumann theorem [1, 6]. This identification is not merely a representation-theoretic convenience. The Weil module carries the canonical symplectic structure associated with the Heisenberg commutator. Consequently, the automorphisms preserving the admissible commutator act through the metaplectic lift:

$$F_n \simeq V_\rho \quad \Rightarrow \quad \mathrm{Mp}(2,\mathbb{R}).$$

The passage to $\mathrm{Mp}(2,\mathbb{R})$ is therefore structural: it is the double cover naturally associated with the symplectic action underlying the Weil representation.

2.2 Lorentzian complexification

The metaplectic lift is initially real and symplectic. The effective geometry closed in the geometric branch is Lorentzian: the effective co-metric established in [16] satisfies

$$g^{\mu\nu} = 2\eta^{\mu\nu}.$$

The relevant spin group is therefore not the compact group $\mathrm{SU}(2)$, but $\mathrm{Spin}(3,1) \simeq \mathrm{SL}(2,\mathbb{C})$.

The Lorentzian closure does not require an independent postulate of a spin group. It complexifies the real metaplectic Lie algebra:

$$\mathfrak{mp}(2,\mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C}). \tag{6}$$

The Lorentzian spin group integrating this complexified algebra is

$$\mathrm{Spin}(3,1) \simeq \mathrm{SL}(2,\mathbb{C}). \tag{7}$$

This defines the admissible spinor bundle S_Π as the associated bundle carrying the fundamental $\mathrm{SL}(2,\mathbb{C})$ spin representation.

2.3 The admissible spin frame bundle

Let P_Π denote the principal $\mathrm{Spin}(3,1)$-bundle obtained from the Lorentzian complexification of the metaplectic lift. The admissible spinor bundle is

$$S_\Pi = P_\Pi \times_{\mathrm{Spin}(3,1)} \mathbb{C}^2.$$

The construction is projective: P_Π is not postulated as an independent spin frame bundle over a pre-existing manifold. It is induced from the Weil fibre through the metaplectic lift and the Lorentzian closure of the effective metric.

2.4 Electroweak tensor sectors

The tensor functors of S_Π define the electroweak bundle data. The symmetric square gives the complex adjoint sector:

$$\mathrm{Sym}^2(S_\Pi) \simeq \mathrm{ad}_{\mathbb{C}}(P_\Pi). \tag{8}$$

Equivalently, fibrewise,

$$\mathrm{Sym}^2(\mathbb{C}^2) \simeq \mathfrak{sl}_2(\mathbb{C}). \tag{9}$$

The Hermitian structure inherited from the admissible Weil module selects the compact real form

$$\mathfrak{su}(2) \subset \mathfrak{sl}_2(\mathbb{C}),$$

and hence the $\mathrm{SU}(2)_L$ gauge sector.

The antisymmetric square gives the determinant line:

$$\wedge^2(S_\Pi) = L_Y.$$

This line is the geometric support of the $U(1)_Y$ factor. It fixes the existence of the hypercharge line, while the distribution of hypercharge weights is constrained by spectral anomaly cancellation in Section 4.

Theorem 2.1 (Spinorial electroweak bundle)**.** *The admissible Weil module V_ρ induces, through its metaplectic lift and the Lorentzian complexification forced by the effective metric [16], an admissible spinor bundle S_Π. Its tensor sectors satisfy*

$$\mathrm{Sym}^2(S_\Pi) \simeq \mathrm{ad}_{\mathbb{C}}(P_\Pi), \qquad \wedge^2(S_\Pi) = L_Y.$$

After selection of the compact real form, these define the $\mathrm{SU}(2)_L \times U(1)_Y$ electroweak bundle data.

Proof. The identification $F_n \simeq V_\rho$ gives the admissible fibre the canonical symplectic structure of the Weil representation [1, 6]. The automorphisms preserving this structure lift to $\mathrm{Mp}(2,\mathbb{R})$. The Lorentzian metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ [16] forces the complexification $\mathfrak{mp}(2,\mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$, whose Lorentzian integration is $\mathrm{Spin}(3,1) \simeq \mathrm{SL}(2,\mathbb{C})$. The associated fundamental bundle is S_Π. The fibrewise identities

$$\mathrm{Sym}^2(\mathbb{C}^2) \simeq \mathfrak{sl}_2(\mathbb{C}), \qquad \wedge^2(\mathbb{C}^2) \simeq \det(\mathbb{C}^2)$$

give the complex adjoint sector and the determinant line. The Hermitian admissibility structure selects the compact real form $\mathfrak{su}(2)$, while the determinant line gives L_Y. □

Fermions arise from the spinorial interpretation of the Weil module after the Lorentzian complexification of its metaplectic Lie-algebraic structure.

3 The Projected Dirac Operator and the Projective Endomorphism

The preceding section constructs the admissible spinor bundle S_Π. We now define the Dirac operator carried by this bundle, isolate the endomorphism induced by non-injective projection, and establish its role in chiral selection.

3.1 The minimal spin–gauge Dirac operator

Let $\nabla^{S,A}$ denote the spin–gauge connection on S_Π, obtained by combining the Levi–Civita spin connection of the effective metric g with the admissible gauge connection A of [4, 18]. The minimally coupled Dirac operator is

$$\mathcal{D}_{g,A} = \gamma^\mu \nabla_\mu^{S,A}. \tag{10}$$

In an ordinary spin^c geometry over a given manifold, the square of this operator is governed by the standard Lichnerowicz formula. In the present setting, S_Π is not a primitive spinor bundle over a pre-existing manifold: it is the spinorial face of the admissible Weil fibre, and the projection Π acts non-trivially on the spinorial sector.

3.2 The projective correction and the definition of E_Π

Let $\Pi_S\colon S \to S_\Pi$ denote the restriction of the admissible projection to the spinorial bundle. The projected Dirac operator is defined by

$$\mathcal{D}_{\Pi,g,A} := \Pi_S \circ \mathcal{D}_{g,A} \circ \Pi_S^*, \tag{11}$$

where Π_S^* denotes the admissible minimal lift, equivalently the adjoint of Π_S after choosing the induced Hermitian structure on the projected spinor bundle. We do not assume that $\Pi_S\Pi_S^* = \mathrm{id}_{S_\Pi}$; this identity holds precisely in the injectively resolved limit and is in general replaced by a non-trivial projector. We assume throughout this section that the admissible lift is symbol-compatible: the principal symbol of $\mathcal{D}_{\Pi,g,A}$ coincides with the projected Clifford symbol $\gamma^\mu\xi_\mu$. Equivalently, the projection defect affects only the zero-order part of $\mathcal{D}^2_{\Pi,g,A}$, not its principal symbol.

Definition 3.1 (Projective endomorphism)**.** The projective endomorphism $E_\Pi \in \Gamma(\mathrm{End}(S_\Pi))$ is defined by

$$E_\Pi := \mathcal{D}^2_{\Pi,g,A} - \left(-(\nabla^{S,A})^2 + \frac{R}{4} + \frac{1}{2}\gamma^\mu\gamma^\nu F^a_{\mu\nu}T^a_R\right). \tag{12}$$

The right-hand side of (12) collects the terms of the standard Lichnerowicz formula. The endomorphism E_Π is therefore not an additional matter potential. It is the internal spectral residue of non-injective projection in the spinorial sector: the part of $\mathcal{D}^2_{\Pi,g,A}$ not resolved by the effective pair (g, A).

Lemma 3.2 (Projective origin of E_Π)**.** *Let S_Π be the admissible spinor bundle induced by the Weil fibre, and assume that the admissible lift Π_S^* is symbol-compatible with the spin–gauge connection. The projected square $\mathcal{D}^2_{\Pi,g,A}$ then differs from the ordinary spin–gauge Lichnerowicz square by a canonical zero-order endomorphism $E_\Pi \in \Gamma(\mathrm{End}(S_\Pi))$. If the spinorial projection is injectively resolved by the effective pair (g, A), then $E_\Pi = 0$. Conversely, $E_\Pi \neq 0$ whenever the non-injective spinorial fibres leave a non-trivial zero-order trace in the projected square. Thus axiom A2 of [1] makes E_Π the natural place where structural non-injectivity enters the spinorial spectral action.*

Proof. When Π_S is injective, Π_S^* is its right inverse and $\Pi_S\Pi_S^* = \mathrm{id}_{S_\Pi}$. The composition (11) then reduces to $\mathcal{D}_{g,A}$ on S_Π, whose square is the standard Lichnerowicz formula; hence $E_\Pi = 0$. When $\Pi_S\Pi_S^* \neq \mathrm{id}_{S_\Pi}$, the squared projected operator contains additional terms controlled by the projection defect. By symbol-compatibility, these terms do not modify the principal Clifford symbol and therefore contribute only to the zero-order part of the Laplace-type operator. They are collected canonically in E_Π via Definition 3.1. □

Remark 3.3. Symbol-compatibility is also required in Section 4: the computation of the Seeley–DeWitt coefficients of $\mathcal{D}^2_{\Pi,g,A}$ via the standard heat-kernel expansion presupposes that the principal symbol is the Clifford symbol, so that the leading heat-kernel term is the standard Gaussian. The hypothesis is therefore not an isolated technical convenience, but a coherence condition between the projective construction and the spectral action formalism.

3.3 Comparison with the Connes–Chamseddine approach

This is the point at which the present construction differs structurally from the Connes–Chamseddine spectral action [19]. In the almost-commutative setting, the finite internal Dirac operator is part of the input data of the spectral triple, and Yukawa matrices and fermion masses are encoded directly in this finite component. Here E_Π is not postulated as a finite Dirac datum: it is induced by the non-injective projection itself. The task is therefore not to choose E_Π, but to determine which endomorphisms are compatible with admissibility, gauge invariance, and spectral coherence. This inversion of logical order is the distinctive feature of the present approach: internal fermionic structure is not input but output.

3.4 Chiral admissibility and the $V - A$ structure

Let $\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ be the chirality operator of $\mathrm{Cl}(T^*M_{\mathrm{eff}}, g^{\mu\nu})$, whose existence as a global section depends on the Lorentzian closure of [16]. Define the chiral projectors

$$P_L = \frac{1}{2}(1-\gamma_5), \qquad P_R = \frac{1}{2}(1+\gamma_5).$$

Definition 3.4 (Chiral projective admissibility)**.** The projective endomorphism E_Π is called *left-admissible* if

$$P_R\, E_\Pi\, P_R = 0. \tag{13}$$

Equivalently, the purely right-handed self-coupling of the projective residue vanishes.

3.5 Spinorial lift of BI parity

The spinorial lift is established here as a theorem rather than assumed. We work on the limiting spinor bundle, on which the Born–Infeld parity of the conjugate Weil sectors is represented by an exact antilinear involution

$$K : V_\rho \longrightarrow V_\rho, \qquad K^2 = 1,$$

derived at the scalar Weil level in [3, 12]; the finite-q error is confined upstream to the convergence of the Weil block toward K, at order $O(q^{-1/2})$ [12, 17]. The two BI-conjugate sectors fix a canonical weight decomposition $V_\rho = V_+ \oplus V_-$, which is not an external Cartan choice but the same datum as the parity: K exchanges the two sectors, and the grading

$$H := 2J_3, \qquad H|_{V_+} = +1, \qquad H|_{V_-} = -1, \qquad H^2 = 1,$$

records which sector a vector belongs to.

Lemma 3.5 (BI parity is off-diagonal with respect to its grading)**.** *The limiting BI involution K and the weight grading H satisfy*

$$HK = -KH. \tag{14}$$

Proof. For $v_+ \in V_+$ one has $Kv_+ \in V_-$, hence $HKv_+ = -Kv_+$ and $KHv_+ = Kv_+$, so $HKv_+ = -KHv_+$. The same computation on V_- gives the operator identity. $\square$

Equation (14) is the operator expression of the fact that the BI parity exchanges precisely the two weights that the same BI datum canonically grades.

Lemma 3.6 (Unique covariant antilinear structure)**.** *After the Lorentzian complexification $\mathfrak{mp}(2,\mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$ of [16], the admissible doublet becomes a Weyl representation $S_L \simeq \mathbb{C}^2$ of $\mathrm{SL}(2,\mathbb{C})$, with $S_R \simeq \overline{S_L}$ inequivalent to S_L. Consequently no Lorentz-covariant antilinear conjugation $S_L \to S_L$ exists, so any covariant antilinear lift must exchange the chiral subbundles $P_L S_\Pi \leftrightarrow P_R S_\Pi$, and the unique $\mathrm{SL}(2,\mathbb{C})$-invariant bilinear form on S_L is, up to scale,*

$$\varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \varepsilon^2 = -1.$$

Proof. A covariant antilinear endomorphism of S_L is a covariant isomorphism $S_L \simeq \overline{S_L}$, that is $\rho \simeq \overline{\rho}$ for the fundamental representation; the $\mathrm{SL}(2,\mathbb{C})$ Weyl representations $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$ are inequivalent, so no such map exists, which forces the off-diagonal exchange. For the form, invariance $g^T B g = B$ under the diagonal torus forces the diagonal entries of B to vanish, and invariance under the unipotent forces the off-diagonal entries to be opposite, so $B \propto \varepsilon$, which is itself invariant under $\mathrm{SL}(2,\mathbb{C})$. Under the compact form $\mathfrak{su}(2)$ the doublet is pseudoreal [7]; the Lorentzian complexification is precisely what removes the equivalence $\rho \simeq \overline{\rho}$ and converts the internal weight exchange into a chiral exchange. $\square$

Theorem 3.7 (Spinorial BI lift). *On the limiting spinor bundle S_Π there exists a unique antilinear lift of the BI parity compatible with both the BI datum (K,H) and $\mathrm{SL}(2,\mathbb{C})$-covariance. Its internal part is the grading-dressed parity $J_\Pi^{\mathrm{int}} = HK$, which in the weight basis ($K = \sigma_x \circ \overline{(\cdot)}$, $H = \sigma_z$) equals the unique covariant form $\varepsilon \circ \overline{(\cdot)}$, since $\sigma_z\sigma_x = \varepsilon$. The lift satisfies*

$$J_\Pi^2 = -1, \qquad J_\Pi\gamma_5 = -\gamma_5 J_\Pi. \tag{15}$$

Proof. By Lemma 3.6 the lift is off-diagonal with internal part proportional to $\varepsilon \circ \overline{(\cdot)}$, and compatibility with the BI datum fixes this internal part to HK. This is Lorentz-legitimate although H alone is not invariant, precisely because HK coincides with the unique invariant $\varepsilon \circ \overline{(\cdot)}$ of Lemma 3.6; the same two constraints determine the lift uniquely. The quaternionic sign is then forced rather than imposed:

$$(HK)^2 = HKHK = -H^2K^2 = -1,$$

using (14), $H^2 = 1$, and $K^2 = 1$. The first identity in (15) follows from $(HK)^2 = -1$; the second from the off-diagonal exchange of the chiral subbundles, which reverses chirality while remaining involutive and therefore contributes no sign to the square. □

The first condition in (15) expresses the quaternionic character of the spinorial lift. The second states that J_Π reverses chirality. Together they imply

$$J_\Pi P_L = P_R J_\Pi, \qquad J_\Pi P_R = P_L J_\Pi, \tag{16}$$

so the BI parity lift exchanges the two chiral subbundles $P_L S_\Pi$ and $P_R S_\Pi$.

Proposition 3.8 (BI parity and left-admissibility). *By Theorem 3.7, and assuming that admissibility selects the orientation-compatible branch, E_Π is left-admissible:*

$$P_R E_\Pi P_R = 0.$$

Proof. By (16), any right-handed self-coupling of a section of S_Π is mapped under J_Π to a left-handed coupling. The admissibility condition selects the branch compatible with the orientation carried by J_Π; equivalently, projective self-couplings of $P_R S_\Pi$ are not orientation-compatible with the BI parity involution. Hence $P_R E_\Pi P_R = 0$. □

The commutator

$$[E_\Pi, \gamma_5] \tag{17}$$

measures the failure of the projective residue to be chirally symmetric. Under the orientation-compatible admissibility branch, a non-trivial projective endomorphism E_Π need not commute with γ_5. The commutator (17) is therefore the projective measure of chiral asymmetry entering both the $V - A$ structure and the spectral anomaly condition of Section 4.

4 Spectral Anomaly Cancellation and Hypercharge Rigidity

The preceding section shows that $\mathcal{D}^2_{\Pi,g,A}$ is a Laplace-type operator with standard Clifford principal symbol and projective endomorphism E_Π. We now use this structure to formulate the hypercharge consistency condition. The guiding principle is that the projected spectral functional $S_\Pi[g, A, \mathcal{D}_{\Pi,g,A}]$ must be invariant under $U(1)_Y$ gauge transformations. In the spinorial sector, this invariance imposes trace constraints on the hypercharge weights carried by the projected components of E_Π. These constraints are the spectral form of anomaly cancellation.

4.1 Projected $U(1)_Y$ variation

Let $L_Y = \wedge^2(S_\Pi)$ be the determinant line identified in Section 2. The projected spinorial sector decomposes into irreducible components $S_R \subset S_\Pi$, each carrying a definite chirality and hypercharge weight Y_R. A $U(1)_Y$ gauge transformation acts by

$$\psi_R \longmapsto e^{iY_R\alpha}\psi_R, \tag{18}$$

and the infinitesimal action is generated by the endomorphism $Y = \bigoplus_R Y_R \,\mathrm{id}_{S_R}$. The projected spectral functional is hypercharge-consistent if

$$\delta_{\alpha Y} S_\Pi[g, A, \mathcal{D}_{\Pi,g,A}] = 0 \tag{19}$$

for all smooth gauge parameters α.

4.2 The γ_5-weighted a_4 coefficient

By symbol-compatibility (Section 3.2), $\mathcal{D}^2_{\Pi,g,A}$ is a Laplace-type operator with Clifford principal symbol, written in standard form as

$$\mathcal{D}^2_{\Pi,g,A} = -(\nabla^{S,A})^2 + E, \tag{20}$$

with

$$E = \frac{R}{4} + \frac{1}{2}\gamma^\mu\gamma^\nu F^a_{\mu\nu} T^a_R + E_\Pi. \tag{21}$$

In four dimensions, the local chiral anomaly is controlled by the γ_5-weighted a_4 coefficient of the Seeley–DeWitt expansion. We denote the corresponding local endomorphism-valued anomaly density by $\mathcal{A}_\Pi$. After taking the spinorial trace, it defines the scalar anomaly density entering the variation of the projected spectral functional. It is determined by the curvature of $\nabla^{S,A}$, the projective endomorphism E_Π, and the effective metric g.

The local chiral anomaly density is

$$\mathcal{A}_\Pi(x) = a_4\big(x, \mathcal{D}^2_{\Pi,g,A}; \gamma_5\big)\,. \tag{22}$$

Lemma 4.1 (Spectral trace form of the anomaly)**.** *Assume symbol-compatibility of the projected Dirac operator. Then the anomalous $U(1)_Y$ variation of the projected spectral functional is*

$$\delta_{\alpha Y} S_\Pi = \int_{M_{\mathrm{eff}}} \alpha\ \mathrm{Tr}_{S_\Pi}(\gamma_5\, Y\, \mathcal{A}_\Pi(x))\, \sqrt{|g|}\, d^4x, \tag{23}$$

where $\mathcal{A}_\Pi$ is the local endomorphism-valued a_4 Seeley–DeWitt density of $\mathcal{D}^2_{\Pi,g,A}$. Gauge invariance therefore requires the local vanishing condition

$$\mathrm{Tr}_{S_\Pi}(\gamma_5\, Y\, \mathcal{A}_\Pi(x)) = 0 \qquad \textit{for all } x \in M_{\mathrm{eff}}. \tag{24}$$

Remark 4.2. The appearance of the γ_5-weighted a_4 coefficient is consistent with the spectral stratification established in the bosonic sector [14]. In the bosonic variation, the a_2 coefficient gives the gravitational response and the a_4 coefficient gives the gauge response. In the spinorial sector, the same a_4 level controls the local chiral anomaly and therefore the hypercharge consistency of the fermionic gauge coupling. Anomaly cancellation is therefore not a new spectral layer: it is the chiral consistency condition of the gauge-level response, located at exactly the same Seeley–DeWitt order as Yang–Mills dynamics. Schematically:

$$a_4 \;\longrightarrow\; \begin{cases} \delta_A S_\Pi \Rightarrow \text{Yang–Mills dynamics}, \\ \delta_g S_\Pi\big|_{a_4} \Rightarrow \text{Yang–Mills stress tensor}, \\ \mathrm{Tr}_{S_\Pi}(\gamma_5\, \delta_{\alpha Y}) \Rightarrow \text{anomaly cancellation}. \end{cases} \tag{25}$$

4.3 Trace constraints on projected hypercharge weights

Only terms in a_4 containing four Clifford matrices survive the γ_5-trace. The gauge-curvature part of (21) contributes the topological density

$$\mathrm{Tr}_{S_\Pi}(\gamma_5 Y\,\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma F_{\mu\nu}F_{\rho\sigma}) \propto \mathrm{tr}_R(YF\wedge F). \tag{26}$$

For the abelian factor this gives the cubic $U(1)_Y$ anomaly; for non-abelian factors it gives the mixed $U(1)_Y \times G^2$ anomalies. The projective endomorphism E_Π contributes additional chiral traces through its γ_5-odd part, equivalently through the components that fail to commute with γ_5. Left-admissibility, $P_R E_\Pi P_R = 0$, constrains these contributions to the orientation-compatible sector. They are therefore absorbed into the same projected anomaly condition, without generating independent constraints beyond those arising from the gauge-curvature sector.

Let $\chi_R \in \{+1, -1\}$ denote the chirality eigenvalue of sector R: $\chi_R = +1$ for left-handed sectors ($S_R \subset P_L S_\Pi$) and $\chi_R = -1$ for right-handed sectors ($S_R \subset P_R S_\Pi$), as determined by the left-admissibility condition of Definition 3.4. The vanishing condition (24) specialises, at leading Seeley–DeWitt order, to the following set of spectral coherence constraints:

$$\sum_R \chi_R\, Y_R\, \dim(R) = 0, \tag{27}$$

$$\sum_R \chi_R\, Y_R^3\, \dim(R) = 0, \tag{28}$$

$$\sum_R \chi_R\, Y_R\, \mathrm{tr}_R(T^a T^b) = 0. \tag{29}$$

Equations (27)–(29) are the gravitational, cubic $U(1)_Y$, and mixed $U(1)_Y \times G^2$ anomaly conditions, respectively. They are not imposed externally; they follow from requiring that the projected spectral functional be well-defined under $U(1)_Y$.

Proposition 4.3 (Spectral anomaly constraints). *Assume symbol-compatibility and left-admissibility of E_Π (Theorem 3.7, Proposition 3.8). Then $U(1)_Y$ invariance of the projected spectral functional imposes the projected anomaly constraints (27)–(29) on the hypercharge weights Y_R. The explicit numerical coefficients of a_4 are collected in Appendix A; only the structural form (26) is required for the derivation of (27)–(29).*

4.4 Hypercharge rigidity

Proposition 4.4 (Hypercharge rigidity from spectral coherence). *Assume that the projected spinorial sectors are precisely those of the admissible matter decomposition (5), with the colour factor unconditional at the pointwise level via the single-frequency BI fingerprint mechanism of [15, 13]. Then $U(1)_Y$ gauge invariance of the projected spectral functional imposes the constraints (27)–(29) on the hypercharge weights Y_R. The allowed assignments are therefore not free parameters of the projected Dirac operator, but spectral coherence data constrained by E_Π.*

Proposition 4.5 (Minimal hypercharge assignment). *If, in addition, the determinant line $L_Y = \wedge^2(S_\Pi)$ carries the minimal integral normalisation compatible with the $U(1)_Y$ action on the projected spinorial sectors, then the constraints (27)–(29) select the Standard Model hypercharge pattern up to an overall sign and rescaling.*

The hypercharge weights are therefore not inserted as independent labels of matter fields. They are the weights of the determinant-line action that survive the spectral coherence condition imposed by the projected Dirac operator. The three-step derivation is:

$$L_Y = \wedge^2(S_\Pi) \;\Rightarrow\; \text{support of } U(1)_Y \;\Rightarrow\; \delta_{\alpha Y} S_\Pi = 0 \;\Rightarrow\; Y_R \text{ rigidity}. \tag{30}$$

5 Generation Multiplicity from the Rank-Three Admissible Invariant

The preceding sections derive the spinorial electroweak bundle and the hypercharge consistency conditions, accounting for the leptonic spinorial sector. We now address the third structural output of the fermionic sector: generation multiplicity. The starting point is the admissible saturation invariant established in the spectral admissibility sub-programme [11],

$$\sigma_c(n_3) = 3. \tag{31}$$

This invariant has already been used in the geometric branch of the programme: under the geometric reconstruction functor it yields the three effective spatial directions (Q5b [2], Q11 [16]). The purpose of the present section is to show that the same invariant admits a distinct spinorial multiplicity reading.

5.1 Two functorial readings of the same invariant

Let $\mathcal{A}_3$ denote the rank-three admissible sector selected at BI saturation:

$$\dim \mathcal{A}_3 = 3. \tag{32}$$

Let $\{e_1, e_2, e_3\}$ be a basis of stable admissible directions in $\mathcal{A}_3$, orthogonal with respect to the admissibility form. There are two distinct functorial readings of $\mathcal{A}_3$.

The geometric reading

$$\mathcal{F}_{\text{geom}} \colon \mathcal{A}_3 \longrightarrow \mathbb{R}^3_{\text{space}} \tag{33}$$

sends each stable admissible direction to a spatial tangent direction. This is the branch used in the reconstruction of the effective Lorentzian metric [2, 16].

The spinorial multiplicity reading is defined as follows.

Definition 5.1 (Spinorial multiplicity functor)**.** The spinorial multiplicity functor $\mathcal{F}_{\text{spin}} \colon \mathcal{A}_3 \to \mathrm{Mult}(S_\Pi)$ assigns to each stable admissible direction $e \in \mathcal{A}_3$ a copy $S_\Pi^{(e)}$ of the admissible spinor bundle, subject to the following conditions:

(i) $S_\Pi^{(e)} \cong S_\Pi$ as $\mathrm{Spin}(3,1)$ spinor bundles;

(ii) the gauge group G_Π acts on $S_\Pi^{(e)}$ via the same representation as on S_Π;

(iii) the admissible projection Π acts on $S_\Pi^{(e)}$ via the same kernel structure as on S_Π;

(iv) distinct directions define orthogonal multiplicity sectors: the corresponding projectors P_e satisfy

$$P_e P_{e'} = 0 \quad (e \neq e'), \qquad [P_e, \rho(g)] = 0 \quad \text{for all } g \in G_\Pi.$$

Condition (iv) is the key constraint: it prevents the three copies from being interpreted as a gauge triplet. They form a multiplicity space on which G_Π acts trivially, rather than a new non-trivial representation of G_Π.

5.2 Gauge-singlet character of the generation space

The generation space $\mathbb{C}^3_{\text{gen}}$ is the indexing space of the functor $\mathcal{F}_{\text{spin}}$, spanned by $\{e_1, e_2, e_3\}$. It is not the adjoint triplet $\mathrm{Sym}^2(S_\Pi)$, which carries the SU(2) adjoint action and belongs to the gauge bundle. By condition (ii) of Definition 5.1, the gauge group acts on each copy $S_\Pi^{(e_i)}$ in the same way as on S_Π; it acts trivially on the indexing labels e_i. Therefore

$$\mathbb{C}^3_{\text{gen}} \;\subset\; \ker\!\big(\mathrm{ad}_{\mathrm{SU}(2)} \oplus Y\big)\,. \tag{34}$$

Theorem 5.2 (Generation multiplicity). *Assume the admissible saturation invariant $\sigma_c(n_3) = 3$ [11] and the spinorial multiplicity functor $\mathcal{F}_{\rm spin}$ of Definition 5.1. Then the fermionic matter bundle carries a three-dimensional gauge-singlet multiplicity factor:*

$$S_\Pi^{\rm fermion} = S_\Pi \otimes \mathbb{C}^3_{\rm gen}, \qquad \mathbb{C}^3_{\rm gen} \subset \ker\big(\mathrm{ad}_{\rm SU(2)} \oplus Y\big)\,. \tag{35}$$

The three-generation structure is the spinorial multiplicity image of the same rank-three admissible invariant that yields the three spatial directions in the geometric branch.

Proof. By (31), $\mathcal{A}_3$ has dimension three and admits a basis of three mutually orthogonal stable admissible directions $\{e_1, e_2, e_3\}$. Applying $\mathcal{F}_{\rm spin}$ yields three copies $S_\Pi^{(e_1)}, S_\Pi^{(e_2)}, S_\Pi^{(e_3)}$. By condition (iv) of Definition 5.1, these copies define orthogonal projective sectors commuting with the G_Π-action, so their direct sum decomposes as

$$\bigoplus_{i=1}^{3} S_\Pi^{(e_i)} \;\cong\; S_\Pi \otimes \mathbb{C}^3_{\rm gen}.$$

The gauge group acts on each factor $S_\Pi^{(e_i)}$ via the spinorial representation and trivially on $\mathbb{C}^3_{\rm gen}$ by condition (ii), giving (34). ☐

Remark 5.3. The equality $\dim \mathbb{R}^3_{\rm space} = \dim \mathbb{C}^3_{\rm gen} = 3$ does not identify the two spaces. They are distinct functorial images of the same admissible invariant $\mathcal{A}_3$: the geometric image carries spatial tangent structure, while the multiplicity image carries no electroweak charge. This distinction prevents the generation triplet from being confused with the adjoint SU(2) triplet $\mathrm{Sym}^2(S_\Pi)$.

6 Dynamic lifting of the static generation degeneracy

Theorem 5.2 establishes the generation factor $\mathbb{C}^3_{\rm gen}$ as the gauge-singlet multiplicity space spanned by the three stable admissible directions $\{e_1, e_2, e_3\}$ of the rank-three sector $\mathcal{A}_3$ ((31), (32)), but leaves the quantitative inter-generation mass splitting open. This section isolates the structural mechanism by which such a splitting can be carried, prior to any amplitude analysis. The dimensionless generation structure is probed by the spectrum of E_Π^2 (Definition 3.1) on $\mathbb{C}^3_{\rm gen}$. The physical mass scale and the amplitude of the splitting are not fixed at this stage; they are deferred to the normalisation of the admissibility cascade.

We use the internal structure of the rank-three sector established in the spectral admissibility sub-programme. In the $\mathrm{End}(H_{\rm eff})$ reading of [8, 12], the admissible sector is the symmetric square

$$\mathcal{A}_3 \;\cong\; \mathrm{Sym}^2(V) \tag{36}$$

of the two-dimensional admissible carrier V (the spin-$\frac{1}{2}$ module of [8]), carrying the Weil weight grading. The orthogonal admissible basis $\{e_1, e_2, e_3\}$ is thereby reorganised into the weight basis $\{e_0, e_+, e_-\}$ of J_3-weights $\{0, +1, -1\}$, with e_0 the central weight-zero direction. This is a structural identification, consistent with [8, 12]; it probes the internal structure of $\mathbb{C}^3_{\rm gen}$ and does not alter its gauge-singlet character (Theorem 5.2). The reality structure is the antilinear $J_\Pi = HK$ of the spinorial BI lift (Theorem 3.7), realising the parity $c \leftrightarrow q - c$, which fixes e_0 and exchanges $e_+ \leftrightarrow e_-$ antilinearly (cf. (16)).

6.1 Static obstruction

Proposition 6.1 (Static obstruction). *Let $C \in \mathrm{End}(\mathbb{C}^3_{\rm gen})$ be a static restriction of E_Π^2 to the generation factor satisfying*

(i) *J_Π-reality, $J_\Pi\, C\, J_\Pi^{-1} = C^\sharp$, and*

(ii) *weight-sector preservation,* $[C, J_3] = 0$.

Then $C_{++} = C_{--}$*: the outer pair* $\{e_+, e_-\}$ *is degenerate, and* C *admits no real splitting of the two light generations.*

Proof. Weight preservation (ii) makes C diagonal in the weight basis, so its doublet block is $\mathrm{diag}(C_{++}, C_{--})$. For the real representative, J_Π-reality (i) reads $PCP = C$, where $P = \left(\begin{smallmatrix} 1&0&0\\ 0&0&1\\ 0&1&0 \end{smallmatrix}\right)$ exchanges $e_+ \leftrightarrow e_-$ and fixes e_0. Conjugating the diagonal doublet block by $P|_{\{e_+,e_-\}} = \left(\begin{smallmatrix} 0&1\\ 1&0 \end{smallmatrix}\right)$ swaps its two entries, so $PCP = C$ forces $C_{++} = C_{--}$. $\square$

Remark 6.2 (Sharpness of the hypotheses). J_Π-reality alone is not sufficient. The condition $PMP = M$ on the doublet admits $M = \left(\begin{smallmatrix} a&b\\ b&a \end{smallmatrix}\right)$, whose eigenvalues $a \pm b$ are split whenever $b \neq 0$: the parity fixes the diagonal entries equal but does not diagonalise the block. The degeneracy requires the static weight-sector preservation (ii) in addition to (i). The equatorial covariance $C_c = \mathrm{diag}(1, \frac{1}{2}, \frac{1}{2})$ of [12] satisfies both hypotheses and is the canonical instance.

6.2 The oriented odd sector

Proposition 6.3 (Oriented odd sector)**.** *Let* $\mathcal{T}_\tau \in \mathrm{End}(\mathbb{C}^3_{\mathrm{gen}})$ *be a lift required to*

(i) *annihilate the central level,* $\mathcal{T}_\tau e_0 = 0$,

(ii) *preserve the outer doublet,* $\mathcal{T}_\tau(\{e_+, e_-\}) \subseteq \{e_+, e_-\}$, *and*

(iii) *be* J_Π*-odd,* $J_\Pi\, \mathcal{T}_\tau\, J_\Pi^{-1} = -\mathcal{T}_\tau$.

Then the admissible lift space is two-dimensional, spanned by the weight generator $J_3 = \mathrm{diag}(0, 1, -1)$ *and the doublet rotation* $R_{\mathrm{mix}} = \left(\begin{smallmatrix} 0&0&0\\ 0&0&1\\ 0&-1&0 \end{smallmatrix}\right)$. *The direction* J_3 *is the unique non-mixing direction producing a real mass split. The Carnot dilation* $D = \mathrm{diag}(2, 1, 1)$ *is* J_Π*-even and therefore excluded from this sector.*

Proof. Conditions (i)–(ii) confine $\mathcal{T}_\tau$ to a real 2×2 doublet block M. Condition (iii) for the real representative is $P_2\, M\, P_2 = -M$ with $P_2 = \left(\begin{smallmatrix} 0&1\\ 1&0 \end{smallmatrix}\right)$, giving $M = \left(\begin{smallmatrix} a&b\\ -b&-a \end{smallmatrix}\right)$, a two-parameter traceless family with $\lambda^2 = a^2 - b^2$. The a-direction is $J_3|_{\mathrm{doublet}}$, with real eigenvalues $\pm a$ and no mixing; the b-direction is R_{mix}, with eigenvalues $\pm ib$ at $a = 0$ and pure mixing. A real split exists for any $|a| > |b|$, but J_3 $(b = 0)$ is the only non-mixing splitting direction. The Carnot dilation satisfies $PDP = D$, hence even, and is excluded. $\square$

Remark 6.4 (Mixing direction). The odd sector contains both a real-split direction and a mixing direction. The real cascade model of Subsection 6.3 activates the former but not the latter; the latter is expected, if present, to require the complex metaplectic phase.

6.3 The cascade bridge

The cascade generator $\partial_\tau \in \mathrm{End}(\mathrm{Sym}^2(V_\rho))$ is the continuum limit of the cascade increment, as established in [16]. We model its restriction to the generation factor by the per-step generator of the oriented metaplectic step $g = S_X(t)\, S_Y(s)$, the ordered product of the two horizontal Heisenberg generators in SL(2) form, lifted to $\mathrm{Sym}^2(V)$ and projected onto the J_Π-odd sector of Proposition 6.3.

Proposition 6.5 (Cascade bridge)**.** *In this model the* J_Π*-odd projection of the cascade generator on* $\mathbb{C}^3_{\mathrm{gen}}$ *is*

$$\Pi_{J_\Pi\text{-}odd}\Big(\partial_\tau\big|_{\mathbb{C}^3_{\mathrm{gen}}}\Big) = \alpha\, J_3 + \mu\, R_{\mathrm{mix}}, \qquad \alpha \neq 0. \tag{37}$$

The real-split coefficient α is non-vanishing already for symmetric shears $t = s$, where it is produced by the order of the product, that is by the non-commutativity $[X, Y] = Z$. Reversal of the cascade order, $g = S_Y(t)\, S_X(s)$, reverses the sign, $\alpha_{\mathrm{rev}} = -\alpha$. The mixing coefficient μ vanishes throughout the real symplectic model.

A supporting diagnostic computation builds the rank-three factor from $V = \mathbb{C}^2$ by Sym^2, takes $\log(g)$ on the lifted representation, and projects its J_Π-odd part onto J_3 and R_{mix} in the Hilbert–Schmidt inner product. It returns $\alpha = +0.156$ at $t = s = 0.40$, with $\alpha_{\mathrm{rev}} = -0.156$ and $\mu = 0$.

6.4 Status and the amplitude problem

The qualitative segment is closed:

$$\text{static obstruction} \;\longrightarrow\; \text{oriented lift} \;\longrightarrow\; \partial_\tau \text{ carries } \alpha J_3, \quad \alpha \neq 0. \tag{38}$$

The splitting of the two light generations is forbidden statically by J_Π-reality together with weight preservation (Proposition 6.1), lives in the two-dimensional J_Π-odd sector (Proposition 6.3), and is carried, in the Q11 diagnostic model, by the oriented cascade generator through its non-zero J_3 projection (Proposition 6.5).

The value of α obtained in the real shear model is not a mass prediction. Only its non-vanishing and its sign reversal under cascade reversal are structural. The amplitude problem is deferred to the normalisation of the admissibility cascade. Concretely, the observed splitting enters through a normalised coefficient ε, not through α itself:

$$\frac{\frac{1}{2} + \varepsilon}{\frac{1}{2} - \varepsilon} = \frac{3}{2}, \qquad \varepsilon = \mathcal{N}_{\mathrm{casc}}\, |\alpha|, \tag{39}$$

and the next step is the derivation of the cascade normalisation $\mathcal{N}_{\mathrm{casc}}$ from the transfer chain $c_\chi \to \delta_{\mathrm{pair}} \to \beta^*$ of the spectral admissibility sub-programme [10, 9], not the matching of the model value $\alpha \simeq 0.156$ to the calibration target $\varepsilon = \frac{1}{10}$.

7 The Colour Sector

The preceding sections derive the spinorial electroweak structure and the generation multiplicity factor, accounting for the leptonic spinorial sector. The quark sector requires, in addition, a colour representation. In the present framework, the colour factor is not introduced by enlarging the spinorial construction itself, but by tensoring the admissible spinor bundle with the colour representation supplied by the admissible gauge-fibre programme [15, 13].

Let V_{color} denote the fundamental colour representation of the admissible SU(3) sector. The quark spinor bundle is defined as

$$S_\Pi^{\mathrm{quark}} = S_\Pi \otimes V_{\mathrm{color}}. \tag{40}$$

The full fermionic matter bundle targeted by the construction is therefore

$$S_\Pi^{\mathrm{matter}} = (S_\Pi \oplus (S_\Pi \otimes V_{\mathrm{color}})) \otimes \mathbb{C}^3_{\mathrm{gen}}. \tag{41}$$

7.1 Status of the colour factor

The construction of S_Π, L_Y, and $\mathbb{C}^3_{\mathrm{gen}}$ does not depend on the colour hypothesis. The factor V_{color} depends on the SU(3) sector, which is unconditional at the pointwise level.

The identification of SU(3) as the colour gauge group follows from $[H\text{-color}]_{\mathrm{pointwise}}$ (exact equality of BFS capacity profiles at finite q), established analytically in [15] via the single-frequency structure of the fingerprint (Proposition 4.23 of O31): each fingerprint vector carries frequency $B = c_1 b_1 + c_2 b_2 + c_3 b_3$, and the orbit dilation $B \mapsto \omega B$ is a bijection of $\mathbb{Z}/q\mathbb{Z}$. The modulation-frequency argument of O32 [13] establishes the weaker $[H\text{-color}]_{\text{eff}}$. The triplet colour module therefore unconditionally defines $V_{\text{color}} \in \text{Rep}(\text{SU}(3))$ exactly at finite q, and the quark sector (40) is an unconditional output of the present paper.

Proposition 7.1 (Colour sector extension)**.** *The admissible colour triplet defines a fundamental* SU(3) *representation* V_{color} *[15, 13]. Then the spinorial matter bundle admits the colour-coupled extension*

$$S_\Pi^{\text{matter}} = (S_\Pi \oplus (S_\Pi \otimes V_{\text{color}})) \otimes \mathbb{C}^3_{\text{gen}}. \tag{42}$$

The first summand is the leptonic sector. The second summand is the quark sector. The generation factor $\mathbb{C}^3_{\text{gen}}$ *remains a gauge-singlet multiplicity space under both summands.*

Remark 7.2. The colour extension does not modify the construction of the admissible spinor bundle S_Π. It adds an independent gauge representation factor, proved unconditional at the effective level [15, 13]. The logical status of the fermionic construction is therefore stratified:

$$\begin{cases} S_\Pi \otimes \mathbb{C}^3_{\text{gen}} & \text{electroweak/leptonic carrier, unconditional,} \\ S_\Pi \otimes V_{\text{color}} \otimes \mathbb{C}^3_{\text{gen}} & \text{quark carrier, unconditional at the pointwise level.} \end{cases} \tag{43}$$

8 Conclusion and Open Directions

This paper completes the fermionic step of the spectral stratification programme. The preceding gauge–gravity synthesis showed that gravity and gauge dynamics arise from different Seeley–DeWitt responses of the same admissible spectral functional. The present work extends this principle to the spinorial sector. Fermions are not introduced as external matter fields added to the admissible bundle. They arise as the spinorial face of the Weil module already forced by non-injective admissibility, revealed through the Lorentzian complexification of its metaplectic lift.

The structural chain established in the paper is

$$\Pi \Rightarrow F_n \simeq V_\rho \Rightarrow \mathrm{Mp}(2,\mathbb{R}) \Rightarrow \mathfrak{mp}(2,\mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C}) \rightsquigarrow \mathrm{Spin}(3,1) \simeq \mathrm{SL}(2,\mathbb{C}) \Rightarrow S_\Pi.$$

The admissible spinor bundle S_Π then generates the electroweak bundle data through its tensor functors:

$$\mathrm{Sym}^2(S_\Pi) \Rightarrow \mathrm{ad}_{\mathbb{C}}(P_\Pi), \qquad \wedge^2(S_\Pi) \Rightarrow L_Y.$$

The first sector gives the complex adjoint, from which the $\mathrm{SU}(2)_L$ triplet is selected by the compact real form. The second gives the determinant line supporting the $U(1)_Y$ factor. This establishes the spinorial electroweak carrier without introducing an independent electroweak matter bundle.

The projected Dirac operator

$$\mathcal{D}_{\Pi,g,A} = \Pi_S \circ \mathcal{D}_{g,A} \circ \Pi_S^*$$

was shown to have a Laplace-type square with standard Clifford principal symbol and a projective zero-order endomorphism E_Π:

$$\mathcal{D}^2_{\Pi,g,A} = -(\nabla^{S,A})^2 + \frac{R}{4} + \frac{1}{2}\gamma^\mu\gamma^\nu F^a_{\mu\nu} T^a_R + E_\Pi.$$

The endomorphism E_Π is the internal spectral residue of non-injective projection in the spinorial sector. It is the locus where chirality, hypercharge coherence, and fermionic internal structure enter the spectral action. In this sense, the analogue of the finite internal Dirac operator is not postulated as additional data: it is induced by the projective construction.

The chiral and hypercharge constraints arise from two complementary uses of E_Π. First, left-admissibility of the projective endomorphism yields the $V-A$ structure:

$$P_R E_\Pi P_R = 0.$$

Second, $U(1)_Y$ invariance of the projected spectral functional imposes the local anomaly-cancellation condition

$$\mathrm{Tr}_{S_\Pi}(\gamma_5\, Y\, \mathcal{A}_\Pi(x)) = 0 \qquad \text{for all } x \in M_{\mathrm{eff}},$$

where $\mathcal{A}_\Pi$ is the γ_5-weighted a_4 Seeley–DeWitt density of $\mathcal{D}^2_{\Pi,g,A}$. Thus anomaly cancellation is not an additional spectral layer. It is the chiral consistency condition of the same a_4 level that yields Yang–Mills dynamics in the bosonic variation.

The third output is generation multiplicity. The rank-three admissible invariant $\sigma_c(n_3) = 3$ has two distinct functorial readings. In the geometric branch, it yields the three effective spatial directions. In the fermionic branch, the spinorial multiplicity functor sends it to a gauge-singlet generation factor:

$$\mathbb{C}^3_{\mathrm{gen}} \subset \ker\big(\mathrm{ad}_{\mathrm{SU}(2)} \oplus Y\big)\,.$$

This generation space is not the SU(2) adjoint triplet $\mathrm{Sym}^2(S_\Pi)$. It is a multiplicity space commuting with the gauge action. The resulting fermionic carrier is

$$S_\Pi^{\mathrm{fermion}} = S_\Pi \otimes \mathbb{C}^3_{\mathrm{gen}}$$

in the electroweak/leptonic sector.

The colour-coupled quark sector is obtained by tensoring with the colour module:

$$S_\Pi^{\mathrm{quark}} = S_\Pi \otimes V_{\mathrm{color}}.$$

This step is unconditional at the pointwise level: $[H\text{-color}]_{\mathrm{pointwise}}$ is established analytically in [15]. Accordingly, the matter carrier addressed by the construction is

$$S_\Pi^{\mathrm{matter}} = (S_\Pi \oplus (S_\Pi \otimes V_{\mathrm{color}})) \otimes \mathbb{C}^3_{\mathrm{gen}},$$

with both summands unconditional at the pointwise level.

The spinorial lift of BI parity, formerly the technical core of the $V-A$ result, is now established: the antilinear conjugation $\pi_{q-c}(v) = \overline{\pi_c(v)}$ lifts to the antilinear endomorphism $J_\Pi = HK$ satisfying (15) (Theorem 3.7). Several open directions nevertheless remain sharply delimited. First, the minimal integral normalisation of the determinant line L_Y must be connected explicitly to the Standard Model hypercharge pattern. The anomaly constraints restrict the allowed weights, while the determinant-line normalisation fixes their scale. Second, the amplitude of the generation splitting remains open: Section 6 fixes the qualitative mechanism, but the coefficient ε must still be derived from the cascade normalisation rather than fitted. Third, possible generation mixing requires the complex metaplectic phase, since the real symplectic cascade model activates the J_3 split direction but not the R_{mix} direction.

A further direction concerns the physical content of E_Π. Since E_Π is the internal spectral residue of non-injective projection, it is the natural place where fermion masses, effective Yukawa couplings, and inter-generation mixing should enter. Section 6 takes the first step on this direction: the inter-generation splitting is statically obstructed on $\mathbb{C}^3_{\mathrm{gen}}$ by J_Π-reality together with weight preservation (Proposition 6.1), lives in a two-dimensional J_Π-odd lift sector (Proposition 6.3), and is carried by the oriented cascade generator through its non-zero J_3 projection (Proposition 6.5). The qualitative mechanism of the generation splitting is thereby fixed; its amplitude and the comparison with the charged-lepton mass hierarchy [10, 9] are among the open directions identified above.

The conclusion of the fermionic step is therefore precise. The Standard Model matter carrier is not postulated as an independent field content. Its spinorial, electroweak, chiral, generation, and colour structures arise from distinct functorial readings of the same admissible Weil fibre. The remaining open problems are specific rigidity problems for E_Π and L_Y.

A Seeley–DeWitt Coefficients for the Projected Dirac Square

Under symbol-compatibility (Section 3.2), $\mathcal{D}^2_{\Pi,g,A}$ is a Laplace-type operator of the form $D = -(\nabla^{S,A})^2 + E$ with E as in (21). The standard Seeley–DeWitt expansion gives [20]

$$a_0(x, D) = (4\pi)^{-2}\,\mathrm{tr}(\mathrm{id}), \tag{44}$$

$$a_2(x, D) = (4\pi)^{-2}\,\mathrm{tr}\big(\tfrac{R}{6}\,\mathrm{id} - E\big)\,, \tag{45}$$

$$a_4(x, D) = (4\pi)^{-2}\,\mathrm{tr}\Big(\tfrac{1}{2}E^2 - \tfrac{1}{6}RE + \tfrac{1}{30}\Delta R\,\mathrm{id} + \tfrac{1}{12}\Omega_{\mu\nu}\Omega^{\mu\nu} + (\text{curvature}^2\ \text{terms})\Big)\,, \tag{46}$$

where $\Omega_{\mu\nu}$ is the curvature of $\nabla^{S,A}$ and the curvature^2 terms are tabulated in [20], Section 4.

The γ_5-weighted variant replaces tr by $\mathrm{tr}(\gamma_5\,\cdot\,)$. Only terms containing four Clifford matrices survive; the gauge-curvature part yields the topological density (26), which controls the anomaly constraints (27)–(29).

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